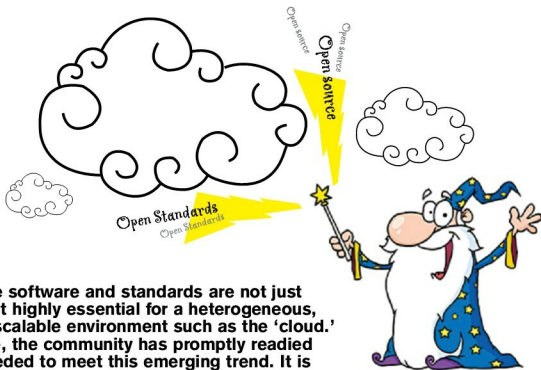


Building the **Cloud** with Open Source and Open Standards



Open Source software and standards are not just beneficial but highly essential for a heterogeneous, shared and scalable environment such as the 'cloud.' What's more, the community has promptly readied the tools needed to meet this emerging trend. It is not surprising that evangelists believe that Open Source has built the cloud...

■ JANANI GOPALAKRISHNAN VIKRAM

Free/Open Source software needs no introduction, but let us clarify what cloud computing really is. Take any definition of cloud computing, and it sounds similar to software-as-a-service (SaaS), utility computing, and even grid computing, to boot. It takes quite a while to figure out the difference, which is why it is best explained to you right away, so you can appreciate the cloud computing concept and the open source advantage even better.

Cloud computing is not a new technology; it is merely a new concept that integrates many virtualisation and pay-as-you-go models—SaaS, platform-as-a-service (PaaS), utility

computing, managed service providers, Web services, and cloud integrators—that already existed. SaaS enables a user to rent and run software applications from service providers who maintain and manage the application on their servers. Remember Salesforce.com. The user simply has to rent an application, use it over the Web, and pay as per the usage.

Utility computing, similarly, enables users to rent infrastructure, such as storage or servers, and use it over the Web. Companies use such services to cater to temporary surges in requirements. Remember IBM, Sun and Amazon.com.

Web services enable developers to connect or fit functional blocks or application programming interfaces

(APIs) offered over the Web into their own applications, so as to not reinvent the wheel. For example: Google Maps and Xignite.

On-demand platforms enable users to string together whatever applications they need from a service provider into a seamless solution and use it over the Internet. This is somewhat like an extension of SaaS, but here the user picks and uses a group of applications, rather than just one. The choice of applications is, however, limited by whatever the service provider has on offer. Remember Force.com, Coghead, and Google AppEngine.

Managed services is also a similar concept; it's just that the user is offered a service—such as network security or backup—over the Web, instead of